# Which Classicality? Incidence, Simplex, and Product-Rule Tests in Finite Quantum Logics

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(Dated: July 13, 2026)

Finite quantum-logical constructions can appear classical or nonclassical depending on which structure is retained. We distinguish incidence tests based on valuations, colorings, and partition representations; simplex-embedding tests for specified prepare-and-measure fragments; and operator-functional tests imposing spectral and product rules. We show that the collective GHZ joint measurement is a single Boolean context and becomes nonclassical only when a common assignment of local factors, together with product preservation, is required. For selected labelled ray fragments, we compute state-depolarizing thresholds for a restricted projector-cone factorization and for a specified vector-generated operational closure, separating exact primal–dual certificates from numerical estimates. These values are properties of the stated fragments and noise model, not invariants of the underlying hypergraphs or a universal ordering of contextuality.

## I. INTRODUCTION

In a single maximal measurement context there is no immediate operational separation between the classical and quantum descriptions of one block. Classically, such a block is a partition, equivalently an equivalence relation on a finite set; quantum mechanically it is represented by a Hilbert-space projection-valued measure (PVM). A Boolean block $C = \{P_1, \ldots, P_d\}$ produces detector-click frequencies $p_i \geq 0$ with $\sum_i p_i = 1$. The same one-context statistics can be read classically as a probability measure on the atoms of a partition, or quantum mechanically as Born probabilities $p_i = \langle\psi|P_i|\psi\rangle$ for a pure state and a rank-one orthogonal context.

Both classical and quantum state spaces are closed under probabilistic mixing: if $P_1$ and $P_2$ are allowed preparations, then $tP_1 + (1-t)P_2$ is an allowed preparation for $0 \leq t \leq 1$. Moreover, the probability of every outcome is the corresponding convex mixture,

$$p(e|tP_1 + (1-t)P_2) = t\,p(e|P_1) + (1-t)\,p(e|P_2).$$

In this precise sense both classical and quantum probabilities are affine under probabilistic mixing. Their state-space geometries nevertheless differ: a classical normalized state space is a simplex, whereas a quantum state space is not, and pure-state transition probabilities are squared projection lengths, for example of Malus type.

For a fixed preparation, even a continuous response curve such as $p(+|\theta) = \cos^2\theta$ can be reproduced by a setting-by-setting classical description: on a common ontic space one may simply stipulate a response function $\xi_{+|\theta}(\lambda)$ whose average has that value. By itself, therefore, the curvature of a one-parameter response is not a simplex obstruction. Such a construction leaves unrestricted how the responses for different settings, and the ontic distributions for different preparations, are related. A nonclassical obstruction arises only when one requires a single model to satisfy additional cross-context constraints: for example, common preparation representations for operationally equivalent mixtures, common response representations for operationally equivalent effects, transformation-composition relations, or Bell locality, where the same local response functions and one shared hidden-variable distribution must work for all choices of local settings. Bell violations rule out precisely such a global Bell-local factorization, not a setting-by-setting fit to an isolated response curve.

The same distinction explains why two contexts, even when pasted along one or more common (intertwining) atoms, do not by themselves produce an operational obstruction. Let two $m$-atom Boolean blocks share a proper subset $S$ of atoms, $0 < |S| < m$. If the two probability assignments agree on every shared atom, then the total probability left for the nonshared atoms is the same in both blocks. That remaining mass can always be coupled by an ordinary joint distribution over predetermined answers for the two measurements. Thus bare two-context statistics, including Firefly-type pastings along a single atom, still have the same operational shadow as a classical marginal model.

A nonclassical probability-level obstruction appears only when a larger family of local distributions is required to be the family of marginals of one global nonnegative distribution, or when a prepare-and-measure table is required to factor through one classical simplex, and that extension or factorization problem becomes infeasible.

Farkas' lemma [1, 2], in the marginal-extension form emphasized by Garg and Mermin [3], concerns the question of whether specified lower-order or context-wise distributions admit compatible higher-order distributions with those marginals. Farkas' alternative says that either such a nonnegative extension exists, or there is a separating linear inequality certifying that no extension exists. Bell [4], Clauser–Horne (CH) [5], and Klyachko–Can–Binicioğlu–Shumovsky (KCBS) [6] inequalities are instances of this general marginal-extension logic; the simplex-embeddability linear program (LP) used below

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is its prepare-and-measure version.

The preceding local observations should therefore not be interpreted as a classical reconstruction of quantum theory. They identify where, and under which limited circumstances, the one-context and (intertwining) two-context shadows may be considered classically reproducible, and where additional operational or marginal constraints must enter.

Once a sufficiently rich family of contexts is retained, the familiar Hilbert-space obstructions reappear. In Hilbert spaces of dimension at least three, quantum propositions cannot in general be assigned pre-existing, context-independent truth values compatible with the functional structure of sharp measurements [7–15]. In quantum-logical language this is first a statement about pasted Boolean algebras, orthomodular posets, or orthogonality hypergraphs. At the operational level it becomes a question about whether a specified prepare-and-measure table admits a classical explanation [3, 6, 16–20]. Generalized-noncontextual ontological models [21], the simplex-embeddability criterion of Schmid, Selby, Wolfe, Kunjwal, and Spekkens [22], and the LP of Selby, Wolfe, Schmid, Sainz, and Rossi [23] give one precise way to formulate that operational question. Operator-valued parity proofs, such as Greenberger–Horne–Zeilinger (GHZ) and Mermin arguments [24–26], retain still another part of the same Hilbert-space description: algebraic relations among observables.

The aim of this paper is to keep these retained structures visible. Incidence data lead to valuation, coloring, and partition-logic questions [12, 27, 28]. Convex-operational data lead to simplex-embedding questions for specified preparations, effects, probabilities, and operational equivalences. Operator data lead to functional-calculus and product-rule questions for assigned observable values. These are related tests, but they are not a hierarchy. The same physical construction can change its classicality status when contexts, effects, local factors, coarse grainings, or operational equivalences are added or forgotten. Our central finding is that classicality is not a binary property of a quantum logical structure, but a bookkeeping choice: which data (incidence, convex-operational, or operator-algebraic) are retained determines which test applies and what obstruction appears.

Five consequences of this bookkeeping guide the paper. A hypergraph can be classically representable as a partition logic while a quantum realization of the same pasting supports nonclassical Born probabilities. A closed fragment consisting of one Boolean context is always simplex-classical. GHZ is not a contradiction within one Boolean joint spectral context; it appears when global product outcomes are identified with products of pre-existing local Pauli values. Simplex embeddability is not the same question as KS uncolorability, and chromatic contextuality is a distinct combinatorial obstruction from ordinary scarcity of two-valued states.

This organization complements existing classifications rather than replacing them. The graph-theoretic approach of Cabello, Severini, and Winter relates exclusivity graphs to invariants such as independence numbers and the Lovász theta function [29]. The hypergraph approach of Acín, Fritz, Leverrier, and Sainz formulates contextuality scenarios directly as compatibility/exclusivity structures with probabilistic models [30]. The sheaf-theoretic framework of Abramsky and Brandenburger classifies empirical models by the obstruction to gluing local sections, distinguishing probabilistic, logical, and strong forms of contextuality [31]. The present distinction is cross-cutting: before applying a classicality test, it asks whether one has retained only incidence data, an operational probability fragment, or operator-algebraic product data.

The numerical computations below therefore use two related fragments. First, a deliberately restricted projector-cone proxy uses the listed rank-one projectors as both preparations and effects, with state-side depolarizing noise. Second, for the rational vector sets, a vector-generated operational closure adds unit effects, residual outcomes of PVMs, complements, coarse grainings, and an optional preparation closure by normalized coarse-grained events [22, 23]. The first audit gives exact-certified benchmarks for several projector cones. The second shows how the numbers change when some of the operational data omitted by the proxy are restored.

The detailed examples are deferred to the body of the paper. GHZ and Peres–Mermin are discussed where operator products and masked contexts are analyzed; meager two-valued-state spaces and chromatic colorability are treated in their own combinatorial sections; and the table of thresholds is interpreted only after the computational protocol has been specified. The introduction therefore only fixes the bookkeeping: which data are being kept, and which classicality test is then being applied.

The paper proceeds as follows. Section II defines the three retained structures. Section III recalls the linear-programming test for simplex embeddability, and Section IV gives its general orthant-factorization interpretation. Section V explains how operator-valued simplex tests depend on whether operators are treated as spectral-outcome data or as algebraic quantities subject to product rules. Section VI explains why two contexts, or weakly pasted contexts, need not by themselves separate vector probabilities from classical convex probabilities. The remaining sections discuss calibration examples, meager two-valued-state spaces, chromatic colorability, the computational protocol, the projector-cone thresholds, and the vector-generated closure comparison.

## II. THREE RETAINED STRUCTURES AND THEIR TESTS

### A. Combinatorial contextuality

Combinatorial contextuality is a property of the pasted context structure. Its input is an orthogonality hypergraph $H = (V, \mathcal{C})$ whose vertices are atoms and whose hyperedges are contexts. The basic question is whether there exist global context-independent assignments compatible with the local Boolean structure. The most familiar assignments are two-valued states

$$s : V \to \{0,1\}, \qquad \sum_{v\in C} s(v) = 1 \quad (C \in \mathcal{C}). \tag{1}$$

Their absence gives the Kochen–Specker (KS) obstruction. Their scarcity gives the meager cases considered below: nonunital, nonseparating, and nonfull two-valued-state spaces. The term *meager* is used here only as a convenient umbrella term for these finite-state-space pathologies; it is not meant in the Baire-category sense. Their abundance, especially a separating family, permits a concrete set representation and hence a partition-logic model [27].

Chromatic contextuality [28] also belongs to the combinatorial level, but it asks for more than the existence of valuations. A strong $d$-coloring of a $d$-uniform hypergraph decomposes the constant-one assignment into $d$ two-valued states, one per color. Thus chromatic contextuality is a spectral-labeling obstruction: there is no way to use one fixed set of $d$ nondegenerate outcome labels globally across all contexts. This is a discrete global obstruction, not a probability-table obstruction.

Partition logics—families of partitions, equivalently families of equivalence relations, on a finite set with generalized urn or automaton realizations [27, 32]—sit at the opposite end of the same combinatorial level. Their existence demonstrates that non-Boolean pasting and complementarity can also be represented on an underlying classical point space. Each partition is the set of equivalence classes of an equivalence relation on the underlying set, and a partition logic is obtained by taking a family of such partitions and identifying equal blocks across different partitions. Its dispersion-free states are the point evaluations determined by the underlying elements of that set. In Wright-style generalized urn models, the same construction is realized by ball types and color filters; in Moore or Mealy automata, it is realized by initial states and input-dependent output partitions. Such models implement complementarity without value indefiniteness [32, 33].

### B. Operational simplex contextuality

Simplex contextuality is an operational convex-geometric notion. Its input is not just a hypergraph but a finite prepare-and-measure fragment

$$(\Omega, \mathcal{E}, p(e|\rho)), \tag{2}$$

where $\Omega$ is a set of preparations, $\mathcal{E}$ is a set of effects, and $p(e|\rho)$ is the observed probability table. In the quantum cases below, $p(e|\rho) = \mathrm{tr}(e\rho)$ and the preparations and effects are rank-one projectors associated with the listed rays.

A finite table by itself is not the nontrivial object: without operational equivalences one can always simulate it by taking one ontic state per preparation and copying the observed response probabilities. The nontrivial question is whether one positive model can extend linearly to the generated preparation and effect cones while respecting the operational equivalences that are part of the fragment.

A fragment is simplex embeddable when its specified pairing factors through a classical simplex. When the unit effect, normalization, and all relevant preparation- and measurement-event equivalences are included, this is equivalent to a generalized-noncontextual model of that operational fragment. For a restricted projector-cone calculation that omits some of those data, it is more precise to call the result a positive cone-factorization benchmark. In either case the property depends on which preparations and effects are included. Enlarging a ray set can change the generated cones and hence the reported robustness, even if the added projectors are an orthogonal completion of an independently specified hypergraph.

The incidence and operational tests can coincide, but need not. A KS set such as the Cabello 18–9 configuration is nonclassical at the valuation level and, for the natural projector fragment, simplex contextual as well. A partition logic, however, can be fully classical at the incidence level while a quantum realization with the same pasting has different Born probabilities.

Conversely, a restricted operational fragment may fail to reveal a defect already present in the underlying hypergraph. Nonunitality is exposed only if the offending atom can actually be prepared or measured; nonseparability only if included preparations statistically distinguish the identified atoms; and a true-implies-false (TIFS) or nonfullness constraint only if the relevant endpoint preparations and effects are present. Without those operational witnesses, the simplex test may remain feasible even though the incidence structure has a combinatorial defect.

### C. Operator-functional contextuality

Operator-functional contextuality, as the term is used here, is not a new rule beyond Kochen–Specker functional composition. It is a separation of that familiar requirement into two parts. The observables are treated as algebraic quantities, not merely as names for their spectral projectors.

The first is *functional-calculus preservation* (FUNC):

$$v(f(A)) = f(v(A)). \tag{3}$$

For a sharp observable with spectral resolution

$$A = \sum_i a_i P_i, \tag{4}$$

this condition is largely equivalent to choosing exactly one value-one spectral projector in the resolved Boolean algebra generated by the $P_i$. When the relevant spectral projectors and their Boolean relations are already part of the combinatorial input, Eq. (3) is therefore mostly reflected in the two-valued-state framework.

The second requirement is *multiplicative preservation* for commuting observables:

$$v(AB) = v(A)v(B), \qquad [A, B] = 0. \tag{5}$$

The product rule is not independent of FUNC when the whole commuting algebra is included: $AB$ is then a function of the joint spectral observable. The operational point here is narrower. After maximal spectral refinement of one context, ordinary projector valuations capture only the Boolean structure of that context, whereas GHZ and Peres–Mermin arguments also identify operator factors across different compatible products. This is the genuinely operator-algebraic ingredient in those parity arguments. It relates different operator presentations that share factors, and it is not captured by looking only at the joint spectral projectors of a single resolved context. Thus the third layer is best understood as having two subcomponents: spectral-functional consistency, which often collapses to projector-valued logic after maximal refinement, and multiplicative consistency, which can impose additional parity constraints across commuting operator products.

This distinction is decisive for GHZ-type arguments. The Mermin–GHZ operators

$$X_1X_2X_3, \quad X_1Y_2Y_3, \quad Y_1X_2Y_3, \quad Y_1Y_2X_3 \tag{6}$$

commute and therefore possess a common eigenbasis. That eigensystem is one orthonormal basis in $\mathbb{C}^8$; as a projector logic it is a single Boolean block with eight atoms and therefore has eight dispersion-free states [26]. There is no KS contradiction at the level of that isolated context. The contradiction enters only when the value of each global product observable is assumed to factor into values of the local Pauli components $X_i$ and $Y_i$. Then each local factor appears twice in the product of the four assigned values, forcing $+1$ classically, whereas the corresponding operator product is $-\mathbb{1}$. The failure is therefore not the absence of a valuation on one context, but the impossibility of a multiplicative, product-preserving valuation on the operator algebra being implicitly invoked. The classical contradiction arises only when one assumes that the value of a global product observable equals the product of pre-existing local values, an assumption that holds in noncontextual ontological models but is operationally unjustified for incompatible local settings.

The Peres–Mermin square [25, 34, 35] is different but related. At the operator level it is a parity proof. When the masked contexts are extracted by simultaneous diagonalization or by matrix pencils, the operator argument can be transcribed into a projective KS structure, in particular its associated 24–24 (vector–context) completion [26, 36] containing the Cabello–Estebaranz–García-Alcaine 18–9 set [13, 26]. Thus some multiplicative operator contradictions descend to combinatorial projector contextuality after unmasking, while others, such as the single-context GHZ eigensystem, do not. This is why the operator-functional layer deserves to be kept separate from both colorability and simplex embeddability, while still recognizing that its functional-calculus part overlaps with the projector-valued combinatorial layer after full spectral resolution.

The word "single-context" is therefore a statement about one particular operational packaging, not about the full GHZ reasoning. The collective measurement of the four global products is a single Boolean context. The local GHZ experiment, however, invokes four incompatible local settings and the counterfactual identification of local factors across them. For the collective packaging the simplex and two-valued-state tests are trivial; for the local-factor packaging the nonclassicality is absolute and rests on locality, noncontextuality, and product preservation. Equivalently, the collective product observable and its local-factor implementation should not be identified without further assumptions. A collective measurement reports a global spectral outcome; a local-factor implementation refines that outcome into local Pauli outcomes and then imposes a product rule on their assigned values.

### D. Probability theories on the same pasting

The probability level is itself not exhausted by the binary alternative "classical or quantum." On a fixed exclusivity structure one may consider at least three types of context-independent additive weights. First, classical probabilities are convex mixtures of dispersion-free states; they form a polytope. Second, Born probabilities arise from a faithful orthogonal representation and a state vector or density operator; graph-theoretically, these are related to Lovász-type theta-body constructions. Third, there are exotic weights, such as Wright-type weights on pentagon-like logics [37, 38], which satisfy additivity within contexts but are neither classical convex-hull points nor Hilbert-space Born probabilities.

Thus the incidence structure alone does not determine the probability calculus. The same pasted logic may admit a partition model, a vector representation, and additional exotic additive weights. This is the sense in which combinatorial contextuality and simplex contextuality are largely independent diagnostics. The combinatorial test classifies global assignments on the logic; the simplex test concerns a particular operational probability model.

## III. THE OPERATIONAL SIMPLEX TEST

Let $\Omega = \{\rho_1, \ldots, \rho_N\}$ be a finite set of quantum preparations, and let $\mathcal{E} = \{e_1, \ldots, e_M\}$ be a finite set of quantum effects. Here each $\rho_i$ is a positive semidefinite preparation operator on $\mathcal{H}$, and each $e_j$ is an effect, $0 \leq e_j \leq I$. The word "state" is therefore used in the operational quantum sense, not in the sense of a two-valued state or valuation on an orthogonality hypergraph.

The preparations may be normalized or subnormalized. Here subnormalized means

$$\rho_i \geq 0, \qquad 0 \leq \mathrm{tr}(\rho_i) \leq 1. \tag{7}$$

Thus one may write $\rho_i = p_i \widehat{\rho}_i$, where $\widehat{\rho}_i$ is a normalized density operator and $0 \leq p_i \leq 1$ is a weight or success probability. This convention is natural for cone constructions, since the cones below are closed under nonnegative rescaling.

Let $\mathcal{H}$ be the finite-dimensional Hilbert space of the fragment, and let

$$\mathcal{A} = \mathrm{Herm}(\mathcal{H}) \tag{8}$$

denote the real vector space of Hermitian operators on $\mathcal{H}$. We do not assume that the chosen coordinates on $\mathcal{A}$ are trace-orthonormal. Instead, choose an arbitrary real basis

$$B_1, \ldots, B_q \tag{9}$$

of $\mathcal{A}$. If

$$A = \sum_{\alpha=1}^{q} a_\alpha B_\alpha, \qquad C = \sum_{\beta=1}^{q} c_\beta B_\beta, \tag{10}$$

then the trace pairing is represented by the Gram matrix

$$G_{\alpha\beta} = \mathrm{tr}(B_\alpha B_\beta). \tag{11}$$

Thus, in these coordinates,

$$\mathrm{tr}(AC) = a^T G c. \tag{12}$$

Only in a trace-orthonormal basis would $G$ be the identity matrix.

The following construction concerns a finite prepare-and-measure fragment: no preparation equivalences beyond normalization, no Bell-locality constraint, and no additional contexts or cross-context relations are imposed.

Let

$$S_\Omega = \mathrm{span}_{\mathbb{R}}(\Omega), \qquad S_E = \mathrm{span}_{\mathbb{R}}(\mathcal{E}) \tag{13}$$

be the accessible preparation and effect subspaces of the real vector space $\mathcal{A}$ of Hermitian operators on $\mathcal{H}$. Choose arbitrary real bases of $S_\Omega$ and $S_E$, and let

$$d_\Omega = \dim S_\Omega, \qquad d_E = \dim S_E. \tag{14}$$

For the factorization statements below, the restricted Born pairing is understood operationally. Thus either it is separating on the chosen accessible spaces, or one first passes to the quotients

$$\overline{S}_\Omega = S_\Omega/(S_\Omega \cap S_E^\perp), \qquad \overline{S}_E = S_E/(S_E \cap S_\Omega^\perp), \tag{15}$$

where orthogonality is with respect to the trace pairing. These quotients identify preparations or effects that the other side of the fragment cannot distinguish. If a quotient is nontrivial, choose linear sections

$$j_\Omega : \overline{S}_\Omega \longrightarrow S_\Omega \subseteq \mathcal{A}, \qquad j_E : \overline{S}_E \longrightarrow S_E \subseteq \mathcal{A} \tag{16}$$

of the quotient maps. If the pairing is already separating, these maps are just the canonical inclusions. Thus the choice of representatives is explicit when it is needed; the induced Born pairing on the quotient spaces does not depend on that choice. To avoid notational clutter, bars are suppressed below, and the dimensions and bases are understood after this reduction. A numerical implementation should verify the separating condition, or perform the indicated quotient, before interpreting the result as an operational simplex test.

After the choices of bases and sections, let

$$I_\Omega \in \mathbb{R}^{q \times d_\Omega}, \qquad I_E \in \mathbb{R}^{q \times d_E}. \tag{17}$$

Concretely, the columns of $I_\Omega$ and $I_E$ are the ambient coordinates, in the basis $B_1, \ldots, B_q$ of $\mathcal{A}$, of the selected representatives of the reduced preparation and effect bases.

Therefore, if $x \in \mathbb{R}^{d_\Omega}$ is the coordinate vector of an operational preparation class and $y \in \mathbb{R}^{d_E}$ is that of an operational effect class, then $I_\Omega x$ and $I_E y$ are chosen ambient representatives. The Born pairing between their classes is

$$\mathrm{tr}(e_y \rho_x) = (I_E y)^T G (I_\Omega x) = y^T I_E^T G I_\Omega x. \tag{18}$$

Thus

$$B_{E\Omega} = I_E^T G I_\Omega \tag{19}$$

is the matrix of the accessible state–effect pairing in the chosen reduced coordinates.

For the original finite fragment, this pairing gives the operational table

$$p_{ij} = \mathrm{tr}(e_j \rho_i), \qquad 1 \leq i \leq N, \quad 1 \leq j \leq M. \tag{20}$$

For normalized preparations these entries are probabilities; for subnormalized preparations they are the corresponding probability weights. Thus the pair $(\Omega, \mathcal{E})$ determines only a finite operational image, or its convex hull, in the corresponding probability space.

Note that a Lovász-theta-type body [39, 40] is not determined by the finite preparation set $\Omega$ alone. It is associated with an exclusivity structure together with an orthogonal representation, or, in the present projective setting, with a fixed family of projective effects carrying the relevant orthogonality relations. Varying the density operator over all quantum states gives the corresponding quantum probability body. Restricting the density operator to the finite set $\Omega$ produces only a finite operational image, or its convex hull, inside that body. The theta-type body describes the quantum probability region associated with an exclusivity structure and an orthogonal representation, whereas the simplex test asks whether a specified finite part of that region factors through a classical simplex. Thus simplex nonembeddability means that the chosen operational fragment lies outside every classical simplex image compatible with the specified preparation and effect cones.

After choosing the reduced bases of $S_\Omega$ and $S_E$, let $V_\Omega$ and $V_E$ denote the matrices whose columns are the coordinate vectors of the $\rho_i$ and $e_j$, respectively. The state and effect cones are first given by their generator, or $V$-, representations:

$$\begin{aligned}\mathrm{Cone}[\Omega] &= \left\{V_\Omega r : r \in \mathbb{R}^N,\ r_i \geq 0\right\} \\ &= \left\{\sum_{i=1}^{N} r_i \rho_i : r_i \geq 0\right\} \subseteq S_\Omega, \qquad (21)\\ \mathrm{Cone}[\mathcal{E}] &= \left\{V_E s : s \in \mathbb{R}^M,\ s_j \geq 0\right\} \\ &= \left\{\sum_{j=1}^{M} s_j e_j : s_j \geq 0\right\} \subseteq S_E. \qquad (22)\end{aligned}$$

Because these cones are finitely generated, they are polyhedral. By the Minkowski–Weyl theorem, every finitely generated polyhedral cone also admits a finite half-space, or $H$-, representation, and conversely [2, 41–46]. This stage is similar to the standard computation of Bell-type inequalities, like the Clauser–Horne–Shimony–Holt inequalities [16, 17]. Those $H$-representations are linear functionals

$$h_1^\Omega, \ldots, h_{n_\Omega}^\Omega \in S_\Omega^*, \qquad h_1^E, \ldots, h_{n_E}^E \in S_E^*, \tag{23}$$

which may be chosen as facet-defining inequalities, such that

$$\mathrm{Cone}[\Omega] = \left\{x \in S_\Omega : h_k^\Omega(x) \geq 0 \text{ for } k = 1, \ldots, n_\Omega\right\}, \tag{24}$$

$$\mathrm{Cone}[\mathcal{E}] = \left\{y \in S_E : h_\ell^E(y) \geq 0 \text{ for } \ell = 1, \ldots, n_E\right\}. \tag{25}$$

Equivalently, collect these functionals as the rows of linear maps

$$H_\Omega : S_\Omega \to \mathbb{R}^{n_\Omega}, \qquad H_E : S_E \to \mathbb{R}^{n_E}, \tag{26}$$

so that

$$H_\Omega x = \begin{pmatrix} h_1^\Omega(x) \\ \vdots \\ h_{n_\Omega}^\Omega(x) \end{pmatrix}, \qquad H_E y = \begin{pmatrix} h_1^E(y) \\ \vdots \\ h_{n_E}^E(y) \end{pmatrix}. \tag{27}$$

With this notation,

$$\begin{aligned} H_\Omega x \geq_{\mathrm{e}} 0 &\iff x \in \mathrm{Cone}[\Omega], \\ H_E y \geq_{\mathrm{e}} 0 &\iff y \in \mathrm{Cone}[\mathcal{E}]. \end{aligned} \tag{28}$$

Here $\geq_{\mathrm{e}}$ denotes entrywise nonnegativity: all components of the corresponding vector must be nonnegative. It is not the positive-semidefinite order on Hermitian operators.

The simplex-embeddability linear program asks whether the Born-pairing matrix (19) factors through the facet descriptions of the state and effect cones. Equivalently, one asks whether there exists an entrywise nonnegative matrix

$$\sigma \in \mathbb{R}^{n_E \times n_\Omega} \tag{29}$$

such that

$$I_E^T G I_\Omega = H_E^T \sigma H_\Omega, \qquad \sigma \geq_{\mathrm{e}} 0. \tag{30}$$

The dimensions are

$$I_E^T G I_\Omega \in \mathbb{R}^{d_E \times d_\Omega}, \qquad H_E^T \sigma H_\Omega \in \mathbb{R}^{d_E \times d_\Omega}. \tag{31}$$

Thus both sides represent the same bilinear pairing between accessible effects and accessible states.

This is the usual simplex embedding written in cone language. The map $x \mapsto H_\Omega x$ sends accessible preparations positively into an orthant, and every positive linear map from a polyhedral cone to an orthant is generated by nonnegative combinations of the facet functionals. Dually, $y \mapsto H_E y$ supplies the positive response functionals. The nonnegative matrix $\sigma$ then couples the two orthants so that the resulting bilinear form agrees with the Born pairing on the accessible quotient spaces.

When Eq. (30) is feasible, the specified pairing admits a positive factorization through a finite orthant. With the unit effect, normalization constraints, operational equivalences, and relevant coarse-grainings included, this is a generalized-noncontextual simplex embedding of the corresponding operational fragment. Without those data it is more accurately described as a positive cone-factorization benchmark.

In particular, if the projector-cone proxy omits $u-e$ for an effect $e$, the LP enforces positivity of the response to $e$, but not necessarily the operational upper bound $\xi_e \leq 1$. The projector-cone thresholds below are therefore not full operational noncontextuality robustnesses. They become instances of the full Schmid–Selby–Wolfe–Kunjwal–Spekkens test only when the operational scenario supplies the unit effect, complements, coarse-grainings, normalization, and the preparation- and measurement-event equivalences at issue.

Equivalently, the cone factorization becomes an ordinary normalized simplex ontological model only after one chooses normalized bases and requires the unit effect to be represented by the all-ones response functional. Then normalized preparations have ontic distributions satisfying $\sum_\lambda \mu_\rho(\lambda) = 1$ and $\xi_u(\lambda) = 1$. In the absence of these order-unit constraints, Eq. (30) should be read as a positive cone-factorization criterion rather than as a full normalized ontological model.

If it is infeasible, one may introduce an operational noise map

$$D : \mathcal{A} \to \mathcal{A} \tag{32}$$

on the state side and minimize $r$ subject to

$$\begin{aligned} I_E^T G\big[(1-r)\,\mathrm{Id}_{\mathcal{A}} + rD\big] I_\Omega = H_E^T \sigma H_\Omega, \\ \sigma \geq_{\mathrm{e}} 0, \qquad 0 \leq r \leq 1. \end{aligned} \tag{33}$$

Here $\mathrm{Id}_{\mathcal{A}}$ denotes the identity map on the ambient Hermitian-operator coordinate space. The optimum $r$ is a robustness against the specified noise. In the calculations below, $D$ is always the completely depolarizing state-side map

$$\rho \longmapsto \mathrm{tr}(\rho)\frac{\mathbb{1}_{d_H}}{d_H}, \tag{34}$$

where $d_H = \dim \mathcal{H}$. Although Eq. (34) is written for positive operators, in Eq. (33) it is used as its unique linear extension to the ambient real vector space $\mathcal{A}$ of Hermitian operators. Accordingly, in the restricted projector-cone audit $r$ is the robustness of the deformed Born pairing for the labelled fragment. It should not be identified with an intrinsic robustness of a larger GPT or laboratory scenario unless the noisy preparations and all associated operational equivalences are also included.

Two qualifications fix the scope of the restricted tests used below. First, the linear program is not a hypergraph-coloring or two-valued-state test. In the projector-cone version, the context structure is seen only through the operator geometry of the listed projectors: orthogonality, linear dependence, and relations such as $\sum_{v\in C} P_v = \mathbb{1}$ when a complete context is present. The program does not impose the deterministic rule that exactly one atom in each block is true.

Second, simplex embeddability is weaker than simpliciality. A generalized probabilistic theory or an accessible fragment may embed into a classical simplex even though its own state space is not itself a simplex. This distinction is central to the generalized-probabilistic-theory analysis of Schmid *et al.* [22]. For accessible fragments, cone equivalence is especially important: cone-equivalent fragments are either both classically explainable or both not, because the embedding problem depends on the generated cones [47].

Finally, Eq. (30) is not the same object as the convex hull of two-valued states of an orthogonality hypergraph. The two-valued-state hull appears as a sharp, deterministic shadow of the operational problem when the fragment contains the eigenstate preparations, sharp projective measurements, and operational equivalences that force deterministic responses on the relevant ontic support. This is the bridge to KS reasoning, not an identity of the two frameworks.

The computations reported in Table I then impose a deliberately restricted *projector-cone* version of the simplex test. The listed rank-one projectors are used as preparations and as effects, and the cone generated by those projectors is tested against state-side depolarizing noise; unit effects, complementary coarse-grainings, and context-induced operational equivalences are not added unless they are generated by the same projector cone. Thus two presentations with the same projector set but different groupings into contexts define the same problem. The thresholds below are consequently benchmarks for these projector cones, not for complete operational scenarios.

## IV. THE ORTHANT-FACTORIZATION CRITERION AND USEFUL DIAGNOSTICS

The exact simplex criterion can be stated compactly as follows. A finite prepare-and-measure fragment is classical at the simplex level exactly when its accessible Born pairing factors through a finite classical orthant. In the coordinate convention of Section III, this Born pairing is the bilinear form

$$(y, x) \longmapsto y^T I_E^T G I_\Omega x. \tag{35}$$

Thus, after quotienting by operational equivalences, there must exist a finite ontic set $\Lambda$ and positive linear maps

$$\mu : S_\Omega \longrightarrow \mathbb{R}_+^\Lambda, \qquad \xi : S_E \longrightarrow \mathbb{R}_+^\Lambda, \tag{36}$$

such that, for all included preparations and effects,

$$\mathrm{tr}(e\rho) = \sum_{\lambda\in\Lambda} \xi_e(\lambda)\mu_\rho(\lambda), \tag{37}$$

with the normalization and response constraints

$$\sum_\lambda \mu_\rho(\lambda) = 1, \qquad \xi_u(\lambda) = 1, \qquad 0 \leq \xi_e(\lambda) \leq 1 \tag{38}$$

for normalized preparations, the unit effect $u$, and included effects $e$. Thus the classical simplex appears as the normalized slice of the orthant $\mathbb{R}_+^\Lambda$. In this language, simplex nonclassicality is precisely the failure of the positive factorization in Eq. (37) for the specified fragment.

The orthant may have dimension larger than the accessible operational space. This possible *dimension gap* is part of the general definition of simplex embeddability [22]. Consequently, the existence of a simplicial cone $K$ satisfying $\mathrm{Cone}[\Omega] \subseteq K \subseteq \mathrm{Cone}[\mathcal{E}]^*$ inside the original accessible state space is a useful sufficient condition, but it is not a necessary reformulation of Eq. (37) unless a no-dimension-gap assumption is added. The nonnegative matrix equation (30), rather than a same-space simplicial sandwich, is the general finite criterion used here. By Farkas duality, infeasibility is witnessed by a separating linear functional, equivalently by a noncontextuality inequality for the chosen operational table.

This geometric picture also explains why the simplex test is not a graph invariant. The orthogonality hypergraph records exclusivity and context incidence, but the simplex linear program sees the accessible cones, the Born bilinear form, and the operational equivalences. Metric information such as $|\langle\psi|\phi\rangle|^2$ matters. Conversely, context grouping is invisible to a projector-only cone test unless it is encoded by additional effects, coarse grainings, or equivalence constraints. The exact question is therefore always whether this specified operational fragment, with these equivalences, admits the positive orthant factorization of Eq. (37).

The following diagnostics are useful, but they should not be mistaken for necessary-and-sufficient graph criteria. First, a fragment whose accessible state cone is already simplicial is normally classically harmless; nonsimpliciality or overcompleteness of the generated cone is a warning sign. But merely having many extreme rays is not by itself an obstruction, because the fragment may still factor through a higher-dimensional orthant. Second, the effects must resolve the nonsimplicial geometry. Coarse or incomplete effects can make a real obstruction invisible. Third, the constraints must close a nontrivial consistency loop: cyclic exclusivity, TIFS or KS implication chains, parity structures after spectral refinement, or equal-mixture preparation equivalences. Without such a loop the hidden-variable sample space often has enough freedom to absorb the transition probabilities.

Three common mechanisms are worth separating. An *overlap or cyclic-exclusivity obstruction* appears when nonorthogonal quantum states and exclusive effects must be reconciled around a closed cycle; the completed pentagon is the canonical low-dimensional calibration. A *determinism obstruction* appears when eigenstate preparations and sharp effects force response functions to take values 0 or 1 on large parts of the ontic space; in sufficiently interlocked configurations this is the operational shadow of KS, TIFS, and related two-valued-state constraints. A *convex-decomposition obstruction* appears when different accessible mixtures represent the same operational preparation and hence must be assigned the same ontic distribution. This last mechanism is the usual preparation-contextuality mechanism and can occur in small scenarios if the relevant mixture equivalences are included.

These mechanisms illuminate the examples below. The firefly logic $L_{12}$ [48] realizes complementarity but does not produce a simplex obstruction in the restricted projector-cone fragment [32, 33]. The completed KCBS pentagon already exhibits a cyclic probability-level gap, although it is not KS uncolorable [6, 38]. Tkadlec's nonunital configuration exposes a deterministic two-valued-state defect [49]. The original Yu–Oh 13-ray set [19] and its 25-ray orthogonal completion used here, the Cabello–Estebaranz–García-Alcaine 18–9 set [13], and the Peres–Mermin square together with the 24–24 completion [25, 26, 34–36] give constrained projector-cone fragments whose Born pairings fail simplex embeddability without noise. GHZ is different: if one retains only the common joint spectral projectors of the four commuting global product observables, the fragment is a single Boolean context and hence simplex-classical. Its nonclassicality enters through the additional requirement that values of global products factor into pre-existing local Pauli values, a product-preservation constraint not present in the projector-only factorization test [24–26].

## V. OPERATOR-FUNCTIONAL TESTS AND MASKED CONTEXTS

An "operator-valued simplex" is not a single unambiguous object. Its meaning depends on what is taken as the operational data and what constraints are imposed on the deterministic classical vertices.

First, if an operator is used only as shorthand for its spectral projectors, then there is no essential change from the projector simplex. A normal observable

$$A = \sum_i a_i P_i \tag{39}$$

contributes expectation values

$$\langle A\rangle_\rho = \sum_i a_i \ \mathrm{tr}(\rho P_i), \tag{40}$$

which are linear post-processings of the spectral-outcome probabilities. If the projectors $P_i$ are already included as effects, adding $A$ itself changes only the coordinates of the data table.

Second, if one keeps only a degenerate operator and does not include enough commuting observables to resolve its eigenspaces, then the simplex test becomes coarser. Some distinctions between projectors are hidden. This can make simplex embeddability easier and may miss contextuality that appears after maximal refinement. The matrix-pencil method addresses precisely this issue: a suitable linear combination of mutually commuting degenerate operators reveals the common spectral projectors and hence the underlying context [26].

Third, if the classical model is required to assign values to operators while preserving functional and product relations, the test changes substantially. The deterministic vertices are no longer arbitrary assignments to spectral outcomes; they must also satisfy relations such as Eq. (5). This is the setting of Peres–Mermin and GHZ parity arguments. In such cases the obstruction may be algebraic even when a projector-only simplex for a single context would be trivial.

Consequently, an operator-valued simplex can mean at least three different things: a simplex for expectation values, a simplex for spectral-outcome probabilities, or a simplex whose vertices are deterministic operator value assignments constrained by functional calculus and, where applicable, by multiplicative product rules. The first two are operational probability tests. The last is an operator-functional test, and the GHZ ingredient is specifically the product-rule part, not merely functional calculus on a single spectral resolution. This is the sense in which GHZ, although a single context after diagonalization, still violates classical expectations: the violation is not in the Boolean algebra of its joint spectral projectors but in the attempted decomposition of its global product observables into context-independent local elements of reality.

### A. Consecutive global measurements versus local factorizations

One might object that the four GHZ product observables commute. Why not simply measure them consecutively on the same three-particle system? In notation, the four observables are

$$\begin{aligned} A_1 &= X_1X_2X_3, & A_2 &= X_1Y_2Y_3, \\ A_3 &= Y_1X_2Y_3, & A_4 &= Y_1Y_2X_3. \end{aligned} \tag{41}$$

This consecutive protocol is possible in principle. Since the $A_i$ commute, their spectral projections commute. An ideal Lüders measurement of one product observable collapses the state only to an eigenspace that is invariant under all the others. Subsequent ideal measurements refine the state to a common eigenspace and preserve the registered eigenvalues. Equivalently, one may perform the joint projective measurement with atoms

$$P_{\boldsymbol{\varepsilon}} = \prod_{i=1}^{4} \frac{\mathbb{1} + \varepsilon_i A_i}{2}, \qquad \varepsilon_i \in \{\pm 1\}, \qquad \varepsilon_1\varepsilon_2\varepsilon_3\varepsilon_4 = -1, \tag{42}$$

where only eight sign patterns are nonzero because $A_1A_2A_3A_4 = -\mathbb{1}$. Only three signs are independent; the fourth is fixed by the product constraint. Thus a collective consecutive protocol registers four repeatable outcomes whose product is always $-1$.

**Remark 1** (Commuting observables and registered outcomes)**.** *The preceding statement uses the standard finite-dimensional spectral fact that commuting self-adjoint observables admit compatible Lüders measurements. Let $A$ and $B$ be self-adjoint operators with $[A, B] = 0$, and let $P_a$ and $Q_b$ be their spectral projectors. Since spectral projectors are obtained from the operators by functional calculus, $[P_a, Q_b] = 0$ for all eigenvalues $a, b$. Hence $BP_a = P_aB$, so the subspace $P_a\mathcal{H}$ is invariant under $B$. Equivalently,*

$$P_a\mathcal{H} = \bigoplus_b P_aQ_b\mathcal{H}, \tag{43}$$

*up to zero summands. Thus the collapse caused by registering the outcome $a$ of $A$ does not cut across the eigenspace decomposition of $B$; it only selects a direct sum of joint eigenspaces.*

*For a Lüders measurement of $A$ followed by a measurement of $B$, the sequential joint probability is*

$$p(a \text{ then } b) = \operatorname{tr}\big(Q_bP_a\rho P_a\big) = \operatorname{tr}\big(P_aQ_b\rho\big).$$

*which is precisely the probability assigned by the simultaneous joint PVM $\{P_aQ_b\}_{a,b}$. A later measurement of $B$ may refine the post-measurement state within $P_a\mathcal{H}$, but it cannot invalidate the registered value $a$. This proof applies to the degenerate observable itself. It does not apply to an arbitrary finer apparatus that, while reporting the same coarse outcome $a$, measures additional degrees of freedom inside $P_a\mathcal{H}$ which need not commute with $B$.*

This collective protocol does not, by itself, yield a GHZ contradiction. It is just one joint measurement, hence one Boolean context. A classical hidden-variable model for it is immediate: the ontic state is one of the eight allowed sign patterns $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4)$ satisfying $\prod_i \varepsilon_i = -1$, and the measurement reads off its four coordinates. No product rule for local factors is used, and no inconsistency follows.

The GHZ contradiction appears only if the same global product values are required to arise from pre-existing local Pauli values, $v(X_i), v(Y_i) \in \{\pm 1\}$, $v(X_1Y_2Y_3) = v(X_1)v(Y_2)v(Y_3)$, with analogous equations for the other three products. Multiplying the four classical equations gives $+1$, because every local factor occurs twice. Quantum mechanically the product of the four commuting global operators is $-\mathbb{1}$. The collective measurement of Eq. (42) reveals the global products, but it never reveals the six local quantities $X_i, Y_i$ whose context-independent coexistence is being assumed. In that precise sense the collective protocol is factor-blind.

The local GHZ protocol operationalizes a different refinement. In each run the three separated parties choose one of the four compatible triples $\{X_1, X_2, X_3\}$, $\{X_1, Y_2, Y_3\}$, $\{Y_1, X_2, Y_3\}$, $\{Y_1, Y_2, X_3\}$, and multiply the local outcomes. These four triples are not jointly measurable by local sharp measurements, since on each particle $X_i$ and $Y_i$ anticommute. Locality, or equivalently the Einstein–Podolsky–Rosen (EPR) counterfactual step in the usual GHZ reasoning, is what motivates assigning the same value to $X_i$ or $Y_i$ independently of which compatible triple is chosen. The contradiction therefore concerns the attempted identification of two operationally different realizations of a product observable: a collective entangled measurement of the product as a whole, and a local measurement of its factors followed by multiplication.

There is a final technical caveat. A degenerate observable does not determine a unique measurement apparatus. A badly chosen implementation may disturb states inside a degenerate eigenspace by effectively measuring additional, possibly incompatible, degrees of freedom.

That is not a failure of commutativity of the abstract observables; it is a different physical refinement of the degenerate measurement. For the ideal consecutive protocol considered here, one assumes Lüders instruments or the common joint PVM, in which case the registered product outcomes are compatible and repeatable. What remains nonclassical is not the sequence of global product outcomes, but the extra product-preserving identification of those outcomes with jointly pre-existing local values.

### B. Single-context triviality, unmasking, and the chromatic analogy

The preceding discussion also fixes a possible misconception about the scope of simplex tests. A closed fragment consisting of one joint spectral measurement is always simplex embeddable. Its ontic states may simply be the atoms of that Boolean algebra, and an arbitrary preparation is represented by the corresponding probability distribution over those atoms. Thus the collective GHZ product fragment, taken by itself, has no projector-simplex obstruction. This is not a positive classical explanation of the full GHZ experiment; it is only the trivial simplex embeddability of a closed single Boolean context.

The qualifier is essential. One should not conclude that operator-valued arguments can never enter a simplex analysis. They can do so whenever the operator proof is unfolded into a multi-context projector fragment, or whenever the operational scenario includes the local factors and their compatibility relations. Peres–Mermin supplies the clearest example: its algebraic parity proof, once the degenerate observables are resolved by simultaneous diagonalization or matrix pencils, yields a genuine multi-context projective configuration, and that configuration can be subjected to two-valued-state and simplex tests. The GHZ collective context does not itself have this property; the local GHZ experiment has a different operational structure, namely four incompatible local settings together with the product-rule identification of shared local factors.

The operative distinction is therefore not whether an argument is operator-valued or simplex-theoretic in the abstract. The right question is whether the retained fragment is merely a single collective spectral context or an unmasked multi-context fragment. A single collective context is classically representable as a Boolean block. A multi-context fragment may display ordinary projector contextuality, simplex nonembeddability, Bell–GHZ nonlocality, or an operator-functional parity contradiction, depending on which structures are retained in the fragment.

This also clarifies the relation to chromatic contextuality. GHZ and strong chromatic contextuality are similar only in the negative sense that neither is exhausted by a probability-table simplex test. Positively they are different obstructions. Chromatic contextuality is combinatorial: it asks whether a fixed nondegenerate spectrum, or equivalently a strong coloring, can be assigned globally across a hypergraph. GHZ is algebraic: it asks whether values can preserve products of commuting observables and their local factors. State-independent parity proofs such as Peres–Mermin are closer in spirit to chromatic obstructions because they express a global spectral inconsistency independent of the input state, but the mechanism is still operator multiplication rather than hypergraph coloring. Thus both belong outside the simplex-probability axis, but they lie on different non-simplex axes: one incidence/coloring-theoretic, the other operator-algebraic.

## VI. WHY TWO CONTEXTS NEED NOT, BY THEMSELVES, REVEAL NONCLASSICALITY

Before turning to calibration examples, it is useful to remove one smaller temptation: the idea that the passage from one context to two already separates quantum vector probabilities from classical convex probabilities. That intuition is correct at the level of the full theories but misleading for a small operational fragment. A single orthonormal context is an ordinary probability simplex. Two nonintertwining contexts are, operationally, a pair of simplexes connected by a stochastic transition matrix.

Let

$$A = \{a_1, \dots, a_d\}, \qquad B = \{b_1, \dots, b_d\} \tag{44}$$

be two orthonormal bases. The quantum transition matrix

$$T_{ji} = |\langle b_j | a_i \rangle|^2 \tag{45}$$

is doubly stochastic. For the restricted fragment consisting only of preparations in $A \cup B$ and measurements in $A \cup B$, this matrix can be reproduced by a classical random channel. One may take ontic states to be pairs $\lambda = (i, j)$, where $i$ is the predetermined answer to context $A$ and $j$ is the predetermined answer to context $B$. Preparing $a_i$ means loading the classical ensemble with the states $(i, 1), \dots, (i, d)$ in proportions $T_{1i}, \dots, T_{di}$. A measurement of $A$ reads the first coordinate and therefore returns $i$ with certainty; a measurement of $B$ reads the second coordinate and returns the Born transition probabilities. Preparations $b_j$ are treated symmetrically. No independence assumption is being imposed on two physical subsystems here; $\lambda = (i, j)$ is only a bookkeeping device for predetermined answers in two otherwise disconnected measurement contexts. The construction works precisely because no further cross-context consistency constraints are present.

If two contexts share an atom, the same pair-coordinate construction must be quotiented so that the shared effect has the same response in both contexts. For weak pastings such as $L_{12}$, this quotienting is exactly what the partition-logic representation accomplishes.

This construction is not a classical simulation of the full Hilbert-space theory, nor of an arbitrary multi-context or Bell scenario. In a Bell experiment a single hidden-variable distribution and a single family of local response functions must work for every local setting; setting-by-setting fits that cannot be assembled into such a common factorization are not Bell-local models. Likewise, additional preparation equivalences, measurement equivalences, linear dependences among preparations or effects, or a third context can already obstruct a common simplex embedding. The limited claim here is only that a bare transition table between two bases, with no further operational constraints beyond normalization, has a classical stochastic-channel realization. Nonclassicality becomes visible when one asks for a single model to satisfy enough cross-context constraints around a nontrivial web of pasted contexts.

This is precisely the lesson of partition logics. A partition logic is obtained by pasting Boolean algebras arising from partitions of a finite set. Its atoms can be represented as subsets of a common classical sample space, and probabilities are ordinary convex mixtures over dispersion-free states. Equivalently, the same structures can be modeled by Wright-style generalized urns or by the initial-state identification problem for finite deterministic Moore or Mealy automata. In a generalized urn, a ball type is an ontic state and different colors correspond to incompatible experimental contexts; looking through one color filter reveals one partition while hiding the others. In the automaton model, the initial state is the ontic state and different input symbols induce different observable partitions of the state set. These models realize complementarity without value indefiniteness.

The contrast with a Hilbert-space realization is then not in the exclusivity graph alone. The same graph may support a set-theoretic partition representation, a faithful orthogonal vector representation, or even more exotic weights [32]. The probability type is determined by the chosen physical representation and the operational fragment, not by the bare pasting diagram alone. Thus the statement "complementarity does not imply contextuality" is not merely philosophical: it is a concrete modeling fact.

## VII. CALIBRATION EXAMPLES: FIREFLY LOGIC AND THE PENTAGON

The first calibration example is the smallest nontrivial pasting. The firefly logic $L_{12}$ is a 5–2 vertex–context set, obtained by pasting two three-atomic Boolean algebras along one common atom [48],

$$\{a,b,c\}, \qquad \{a,d,e\}. \tag{46}$$

A faithful qutrit realization is

$$a=(1,0,0), \quad b=(0,1,0), \quad c=(0,0,1), \tag{47}$$

$$d=(0,1,1), \quad e=(0,1,-1). \tag{48}$$

The partition-logic representation is even more elementary. On the finite set $S_5=\{1,\dots,5\}$ one may use the two partitions

$$\big\{\{2,3\},\{4,5\},\{1\}\big\}, \qquad \big\{\{1\},\{3,5\},\{2,4\}\big\}, \tag{49}$$

which paste along the singleton atom $\{1\}$. This is a generalized urn or finite-automaton realization of the same logic. It is non-Boolean as a pasted structure, but it is still classically representable. Its role here is diagnostic: it is the first place where contexts are not simply disjoint, yet simplex nonembeddability should not be expected merely from that fact.

The next calibration example is the pentagon, or pentagram logic. In the KCBS form one starts with five qutrit rays $v_i$ satisfying cyclic orthogonality

$$v_i \perp v_{i+1} \qquad (i \bmod 5). \tag{50}$$

A convenient set of unnormalized ray representatives is

$$v_k \propto \left(\cos\frac{4\pi k}{5},\ \sin\frac{4\pi k}{5},\ \sqrt{\cos\frac{\pi}{5}}\right), \qquad k=0,\dots,4. \tag{51}$$

Completing each orthogonal pair by $w_i=v_i\times v_{i+1}$ gives five complete qutrit contexts

$$\{v_i,w_i,v_{i+1}\}, \qquad i=0,\dots,4, \tag{52}$$

and hence a 10-ray pentagon (10–5 vertex–context set).

The pentagon is still partition-logically representable: the five cyclically intertwined contexts support 11 two-valued states, from which a partition logic over $S_{11}=\{1,\dots,11\}$ can be reconstructed; explicitly, one may take the 11 admissible two-valued states as the underlying points and represent each atom by the subset of states assigning value one to it. Nevertheless, the pentagon already displays a difference between classical convex probabilities and Born/Lovász-type vector probabilities. On the five outer atoms, the classical convex hull yields the familiar pentagon bound $\sum_i p_i \le 2$, whereas maximization over quantum states for the symmetric orthogonal representation gives $\sqrt{5}$. The numerical threshold reported below belongs to the symmetric algebraic realization in Eq. (51); it is not a graph invariant of $C_5$ alone. This is the first useful calibration point at which cyclic exclusivity exposes a quantitative difference, though still not a KS impossibility.

## VIII. COMBINATORIAL CONTEXTUALITY: TWO-VALUED STATES AND MEAGERNESS

Let $H=(V,\mathcal{C})$ be a finite $d$-uniform orthogonality hypergraph. The vertices $v\in V$ represent atoms, usually rank-one projectors, and each context $C\in\mathcal{C}$ is a $d$-element set of mutually orthogonal atoms corresponding to a maximal sharp measurement. A faithful orthogonal representation in $\mathbb{C}^d$ or $\mathbb{R}^d$ assigns to every vertex a ray $|v\rangle$ such that vertices are orthogonal precisely when the hypergraph says they are, and every context resolves the identity:

$$\sum_{v\in C}\Pi_v = \mathbb{1}, \qquad \Pi_v = \frac{|v\rangle\langle v|}{\langle v|v\rangle}. \tag{53}$$

**Definition 1** (Two-valued state)**.** *A two-valued state on $H$ is a map $s : V \to \{0,1\}$ such that*

$$\sum_{v\in C} s(v) = 1 \qquad \text{for every } C \in \mathcal{C}. \tag{54}$$

*We denote the set of all such states by $\mathcal{S}_2(H)$.*

The deterministic valuation polytope is

$$\mathsf{P}_2(H) = \operatorname{conv}\mathcal{S}_2(H) \subseteq [0,1]^V. \tag{55}$$

It is useful to distinguish several ways in which $\mathcal{S}_2(H)$ can be deficient. Tkadlec called attention to hypergraphs whose state spaces are empty, not unital, not separating, or not full [49]; these notions are standard in the quantum-logic literature [33, 50, 51].

**Definition 2** (Unital, separating, full)**.** *Let $\mathcal{S}_2(H)$ be the set of two-valued states on $H$. It is called*

1. unital *if for every vertex $a \in V$ there exists $s \in \mathcal{S}_2(H)$ with $s(a) = 1$;*
2. separating *if for every distinct pair $a, b \in V$ there exists $s \in \mathcal{S}_2(H)$ with $s(a) \neq s(b)$;*
3. full *if for every nonorthogonal pair $a, b \in V$ there exists $s \in \mathcal{S}_2(H)$ with $s(a) = s(b) = 1$.*

*In the usual orthomodular-poset formulation, where atoms sit inside a poset with zero, unit, and orthocomplementation, fullness implies separation and separation implies unitality. In the atom-only hypergraph formulation used here these implications require the corresponding reconstruction assumptions. We therefore treat unitality, separation, and fullness as separate diagnostics and do not rely on the hierarchy without those assumptions.*

**Definition 3** (Meager two-valued-state space)**.** *A hypergraph has a meager two-valued-state space if $\mathcal{S}_2(H)$ is empty, nonunital, nonseparating, or nonfull. This is an umbrella term; the four cases have different operational meanings.*

Each failure mode corresponds to a different linear constraint on $\mathsf{P}_2(H)$. If $\mathcal{S}_2(H) = \emptyset$, then $\mathsf{P}_2(H) = \emptyset$. This is the KS case. If $\mathcal{S}_2(H)$ is nonunital at $a$, then

$$x_a = 0 \qquad \text{for all } x \in \mathsf{P}_2(H). \tag{56}$$

If it is nonseparating for $a \neq b$, meaning $s(a) = s(b)$ for all $s \in \mathcal{S}_2(H)$, then

$$x_a = x_b \qquad \text{for all } x \in \mathsf{P}_2(H). \tag{57}$$

If it is nonfull for a nonorthogonal pair $a, b$, meaning no $s$ has $s(a) = s(b) = 1$, then

$$x_a + x_b \leq 1 \qquad \text{for all } x \in \mathsf{P}_2(H). \tag{58}$$

A true-implies-false (TIFS) structure, such as the Specker bug, singles out an ordered input–output pair $a, b$ for which

$$s(a) = 1 \Longrightarrow s(b) = 0, \tag{59}$$

for all admissible $s$. At the level of binary assignments this is equivalent to the nonfullness inequality $s(a) + s(b) \leq 1$; the direction enters through the distinguished preparation or conditioning on $a$, not through a stronger binary constraint. Equation (59) implies that the face $x_a = 1$ of the two-valued-state polytope is contained in $x_b = 0$.

## IX. HOW MEAGERNESS BECOMES OPERATIONAL

The preceding notions are combinatorial: they concern the possible two-valued states on a hypergraph. To become operational, however, such defects must be visible in the actual preparations and effects included in the fragment. A logical obstruction that is present in the hypergraph may remain invisible if the fragment is too poor to test it.

Let $H$ have a faithful orthogonal representation, with vertex projectors $\Pi_v$. For a preparation $\rho$, write

$$p_v^\rho = \operatorname{tr}(\rho\Pi_v) \tag{60}$$

for the Born probability assigned to the vertex effect $\Pi_v$. The deterministic two-valued-state polytope

$$\mathsf{P}_2(H) = \operatorname{conv}\mathcal{S}_2(H) \tag{61}$$

contains exactly the probability assignments obtainable as classical mixtures of admissible two-valued states. Thus, if an accessible Born vector $p^\rho$ lies outside $\mathsf{P}_2(H)$, then no deterministic two-valued-state model can reproduce that part of the fragment.

The different forms of meagerness become visible in different ways. If the two-valued states are nonunital at an atom $a$, then every admissible two-valued state assigns

$$s(a) = 0. \tag{62}$$

Consequently every classical mixture in $\mathsf{P}_2(H)$ also assigns probability zero to $a$. But if the fragment allows the eigenstate preparation $\rho = \Pi_a$ and the effect $\Pi_a$, quantum theory gives

$$\operatorname{tr}(\Pi_a\Pi_a) = 1. \tag{63}$$

The atom that is never true in the two-valued-state model is obtained with certainty in the corresponding quantum eigenstate. Nonunitality is then directly operationally visible.

If the two-valued states are nonseparating for two distinct atoms $a$ and $b$, then every admissible valuation identifies them:

$$s(a) = s(b). \tag{64}$$

Every classical mixture therefore satisfies $x_a = x_b$. This becomes operationally relevant only if the Born statistics can distinguish the two effects, that is, if the fragment contains some preparation $\rho$ such that

$$\mathrm{tr}(\rho\Pi_a) \neq \mathrm{tr}(\rho\Pi_b). \tag{65}$$

In that case the classical valuation model merges two outcomes that the quantum statistics separate. If no such preparation is included, the nonseparability remains present combinatorially but is not witnessed by the fragment.

Finally, a true-implies-false (TIFS) or nonfullness constraint becomes visible through nonorthogonal overlap. Suppose $a$ and $b$ are nonorthogonal, but the two-valued states force

$$s(a) = 1 \Longrightarrow s(b) = 0. \tag{66}$$

Classically, conditioning on $a$ being true then forces $b$ to be false. Quantum mechanically, however, the eigenstate $\Pi_a$ gives

$$\mathrm{tr}(\Pi_a\Pi_b) = |\langle a|b\rangle|^2 > 0 \tag{67}$$

whenever $a$ and $b$ are nonorthogonal. Thus, if the fragment contains the preparation $\Pi_a$ and the effects $\Pi_a, \Pi_b$, the TIFS implication is operationally contradicted by the nonzero transition probability from $a$ to $b$.

The qualification is essential. These are direct witnesses against models whose ontic states are the admissible deterministic two-valued states. They become witnesses against generalized-noncontextual models only when the operational scenario justifies the corresponding outcome-determinism and equivalence assumptions for the sharp effects. If a nonunital atom is never prepared or measured, if a nonseparated pair is never statistically distinguished, or if the endpoints of a TIFS gadget are absent from the preparation/effect list, then the combinatorial defect need not appear in the operational table. The noisy cone-factorization LP then quantifies the robustness of the stated fragment, rather than automatically that of every operational realization of the hypergraph.

## X. RELATION TO GENERALIZED NONCONTEXTUALITY

The deterministic two-valued-state analysis is closest to the traditional KS setting. Generalized noncontextuality is broader and does not assume determinism from the outset. For sharp projective measurements, the relevant deterministic constraints can be derived only under the appropriate ideal operational assumptions, including the projective measurements and the preparation or mixture equivalences used in the derivation [52]. The valuation polytopes below should therefore be read as deterministic shadows of suitably specified operational models, not as a substitute for stating those assumptions.

For a fixed accessible GPT fragment with all operational equivalences already quotiented out, the factorization Eq. (30) is the corresponding cone form of the simplex embedding problem. The computations in this paper choose a smaller fragment, so the resulting $r$-values are robustnesses of that fragment, not automatic lower or upper bounds on a separately specified full operational robustness. The closure computation in Table II is one intermediate audit: it restores the unit, residual sharp-measurement outcomes, complements, coarse grainings, and some preparation equivalences generated by the same vector data. It is still not a claim to enumerate every possible laboratory realization of the scenario.

In this situation the two-valued-state polytope $\mathsf{P}_2(H)$ is the deterministic shadow of the operational simplex model. Meagerness of $\mathcal{S}_2(H)$ supplies faces or equalities that the Born vectors may violate. Depolarizing noise moves the Born vectors toward a more symmetric point, and the noisy simplex linear program computes the amount of noise needed to make the entire fragment simplex-embeddable.

The crucial qualification is that the simplex linear program is not merely "the fractional coloring polytope." Its variables are the entries of $\sigma$ in Eq. (30), indexed by facets of the state and effect cones. The number of such facets need not equal the number of vertices or contexts of the hypergraph. In the Yu–Oh 13-ray computation below, there are 13 projectors but 24 facets of each cone, hence $\sigma$ is $24 \times 24$.

## XI. COMBINATORIAL CONTEXTUALITY: CHROMATIC COMPLETENESS

Chromatic contextuality imposes a different global requirement. Let $H = (V, \mathcal{C})$ be $d$-uniform. A strong $d$-coloring is a map

$$c : V \to \{1, \dots, d\} \tag{68}$$

such that every context contains every color exactly once. Equivalently, for every color $k$ define

$$s_k(v) = \begin{cases} 1, & c(v) = k, \\ 0, & c(v) \neq k. \end{cases} \tag{69}$$

Then each $s_k$ is a two-valued state and

$$\sum_{k=1}^{d} s_k(v) = 1 \qquad \text{for every } v \in V. \tag{70}$$

Thus chromatic colorability asks not merely for two-valued states, but for a complete decomposition of the unit assignment into $d$ admissible two-valued states. The obstruction $\chi_s(H) > d$, where $\chi_s$ is the strong chromatic number, means that no such global spectral-label decomposition exists.

This explains why chromatic contextuality is independent of ordinary valuation scarcity. A hypergraph may possess a separating or even full set of two-valued states and nevertheless fail Eq. (70). Conversely, a KS hypergraph with no two-valued states is of course chromatically contextual, but the chromatic description is then not the most discriminating one: the stronger valuation obstruction is already present.

## XII. COMPUTATIONAL PROTOCOL

The computational analysis fixes one convention for all examples. The listed rays define rank-one projectors, these projectors are used both as preparations and as effects, and depolarizing noise is applied only on the state side. The named hypergraphs determine the projector sets and are used in the accompanying combinatorial analysis; in the linear program, their blocks enter only through the operator geometry of the included projectors.

The computations use the non-orthonormal coordinate convention described in Section III. For exact rational audits the program represents real symmetric projectors in the raw coordinate basis

$$(\rho_{11}, \ldots, \rho_{dd}, \rho_{12}, \rho_{13}, \ldots, \rho_{d-1,d}), \tag{71}$$

rather than in a trace-orthonormal basis. In this basis the trace pairing is represented by the diagonal matrix

$$G = \operatorname{diag}\Big(\underbrace{1, \ldots, 1}_{d}, \underbrace{2, \ldots, 2}_{d(d-1)/2}\Big), \tag{72}$$

because the $d$ diagonal entries contribute once whereas the $d(d-1)/2$ off-diagonal real symmetric entries contribute twice to the trace inner product. Thus, if $e$ and $\rho$ denote raw coordinate vectors, then

$$\operatorname{tr}(e\rho) = e^T G \rho. \tag{73}$$

The consolidated audit stores the dual effect vector $\widetilde{e} = Ge$, whose ordinary dot product with $\rho$ gives the same Born probability. This convention avoids square-root factors for integer rays and makes the rational cases suitable for exact polyhedral conversion with `pycddlib`/cddlib over $\mathbb{Q}$.

The audit uses the accessible-fragment formalism. Starting from full raw state coordinates $V_\Omega^F$ and full dual-effect coordinates $\widetilde{V}_E^F$, Stage 1 replaces them by accessible coordinate matrices $V_\Omega^A$ and $\widetilde{V}_E^A$ of full row rank, together with embedding matrices $I_\Omega$ and $\widetilde{I}_E$. When the operational quotient is nontrivial, these matrices represent the chosen linear sections; otherwise they are the ordinary inclusions. In either case they embed the chosen accessible representatives back into the full raw coordinate space, so that

$$V_\Omega^F = I_\Omega V_\Omega^A, \qquad \widetilde{V}_E^F = \widetilde{I}_E \widetilde{V}_E^A \tag{74}$$

up to the chosen basis convention. Stage 2 computes facet descriptions $H_\Omega$ and $H_E$ of the accessible cones. Because the metric has already been absorbed into the dual-effect coordinates, Stage 3 solves

$$\widetilde{I}_E^T\big[(1-r)\,\mathrm{Id}_{\mathcal{A}} + rD\big] I_\Omega = H_E^T \sigma H_\Omega,\ \sigma \geq_{\mathrm{e}} 0,\ 0 \leq r \leq 1. \tag{75}$$

This is Eq. (33) after the accessible reduction and the substitution $\widetilde{I}_E = G I_E$. Stating this substitution explicitly prevents the trace metric from being counted twice.

Candidate optima are found numerically. For rows reported as exact optima, the proposed values are then checked symbolically with rational $H_\Omega, H_E, r$, and $\sigma$, and exact rational dual witnesses are verified.

The certificate status in Table I distinguishes two cases. "Exact optimum" means that both an exact primal certificate and an exact dual certificate have been verified. "Numerical" means that only a floating-point solution and its residual or dual-feasibility check are available. The completed pentagon is algebraic rather than rational in the chosen coordinates and is therefore treated by a numerical branch. The Specker-bug coordinates belong to $\mathbb{Q}(\sqrt{2})$, but their Born table admits an exact rational rank factorization; exact primal and dual witnesses certify $r = 25/79$. The Kochen–Specker $\Gamma_3$ Born table is genuinely quadratic and is treated numerically. The Tkadlec nonunital row has rational coordinates and exact facets, but no exact primal–dual pair was recovered at its numerical optimum.

For the numerical rows, the printed digits are descriptive rather than certified and should be read together with the residual information in Appendix A.

The symmetric choice used in this audit,

$$\Omega = \mathcal{E} = \{\Pi_v : v \in R\},$$

is not part of the simplex-embeddability criterion itself. The general criterion allows the preparation and effect cones to be different. The present choice is a benchmarking convention: every listed ray is assumed to be both sharply preparable and sharply measurable, so that the Born table contains all projector overlaps $\operatorname{tr}(\Pi_v \Pi_w)$. If instead one used only a single orthonormal basis of preparations, the resulting fragment would usually be simplex-embeddable. Indeed, for preparations $\rho_k = |k\rangle\langle k|$ and arbitrary projective effects $\Pi_v = |v\rangle\langle v|$, the classical model with ontic states $\lambda = k$, preparation distributions $\mu_k(\lambda) = \delta_{\lambda k}$, and response functions $\xi_v(\lambda) = |\langle \lambda | v\rangle|^2$ reproduces

$$\operatorname{tr}(\Pi_v \rho_k) = \sum_\lambda \xi_v(\lambda)\mu_k(\lambda).$$

Thus such a preparation-restricted fragment cannot by itself witness the valuation scarcity of the underlying ray hypergraph.

The second audit uses the same vector coordinates but replaces the projector-only effect set by a vector-generated operational closure. For every mutually orthogonal subset of the listed rays, the program forms the sharp measurement consisting of those rank-one projectors together with the residual effect $\mathbb{1}-\sum_i \Pi_i$, whenever this residual is nonzero. It then adds the unit effect and all nonempty proper coarse grainings, and identifies equal operators in the raw coordinate quotient. Two preparation variants are reported. The *closed-effects* variant keeps the original eigenstate preparations and adds the maximally mixed state. The *closed-effects plus preparations* variant also adds normalized versions of the closed measurement events as preparations whenever their trace is nonzero. These closure rows are computed from exact rational facets when possible, but the values in Table II are numerical unless an exact primal and dual certificate is explicitly stated. For the large $\Gamma_3$ and Tkadlec nonunital closure rows, the linear program was solved in a cutting-dual form: this gives the same numerical optimum but does not produce a primal $\sigma$ certificate.

All calculations reported here are produced by one self-contained program, `simplex_audit_rebuilt.py`. Before constructing any cone or LP, it checks projective uniqueness, enumerates all orthogonal contexts, and verifies the advertised valuation properties. The source-derived Specker bug has 13 rays, seven triads, 14 two-valued states, and the distinguished nonfull pair. The $\Gamma_3$ construction of Kochen and Specker [12], in the coordinate-labelled realization of Tkadlec [49], is completed to 27 rays and 17 triads; its 24 two-valued states identify the distinguished nonseparating pair. The Tkadlec nonunital construction starts from the 25 marked rays, completes every orthogonal pair by its unique cross-product ray, and verifies 37 rays, 26 triads, eight two-valued states, and eight never-true rays. Cabello's thesis gives an independent parametric coordinatization of the Bell/Specker TIFS family [**?** ]. The Peres–Mermin rays are checked against the six commuting operator contexts of the square. The machine-readable report stores the vectors, contexts, cone generators, facets, LP matrices, numerical solutions, and all certificates.

OpenAI Codex with a GPT-5.6-based model (July 2026) assisted in consolidating and reviewing the audit code and in revising the exposition. The author specified the mathematical constructions, inspected the generated code, executed all reported runs, and checked the outputs. Exact claims are accepted by the program only after exact rational primal and dual verification; otherwise they remain explicitly numerical.

## XIII. EXAMPLES AND RESULTS

### A. Calibration computations: $L_{12}$ and the completed pentagon

The firefly logic $L_{12}$ is included as a calibration for complementarity without simplex obstruction. Its five projectors span a four-dimensional accessible state space inside the six-dimensional real symmetric qutrit trace space. After the Stage 1 accessible-fragment reduction the exact rational audit finds five facets for each accessible cone and returns

$$r_{L_{12}} = 0, \tag{76}$$

with both exact primal and exact dual certificates. Thus $L_{12}$ is not merely a lower-rank exception to the full-rank trace-space code; it is an exactly simplex-embeddable accessible fragment. This result is consistent with the partition representation of its incidence structure, but it does not follow from that representation alone: a partition-logically representable pasting can support a different quantum Born table, as the pentagon comparison illustrates.

The completed pentagon contains 10 projectors and spans the full six-dimensional real symmetric qutrit trace space. The computation verifies adjacent orthogonality, obtains five contexts of the form Eq. (52), finds 12 cone facets, and returns

$$r_{C_5} \simeq 0.119140568134. \tag{77}$$

The value is numerical only in the present audit because the coordinates involve algebraic quantities outside the rational cddlib backend. It is a positive value for this particular projector-cone proxy; it should not be read as a strength ranking against the later configurations.

### B. Specker bug and the Kochen–Specker $\Gamma_3$ configuration

The Specker bug is represented by 13 rays in seven triads. The coordinate realization uses entries in $\mathbb{Q}(\sqrt{2})$ and is independently supported by the Bell/Specker parametrization in Cabello's thesis [**?** ] and by the coordinate-labelled diagram of Tkadlec [49]. Exact enumeration gives 14 two-valued states and verifies the distinguished nonfull pair, equivalently the TIFS implication after conditioning on its input ray. Although the ray coordinates are quadratic, every Born overlap belongs to $\mathbb{Q}$. The program therefore constructs an exact rational rank factorization of the Born table. The state and effect cones each have 26 facets, and exact primal and dual certificates give

$$r_{\mathrm{bug}} = \frac{25}{79} \simeq 0.316455696203. \tag{78}$$

The configuration denoted $\Gamma_3$ in the Kochen–Specker construction [12] is evaluated using the coordinate-labelled realization recorded by Tkadlec [49]. Orthogonal completion gives 27 rays in 17 triads. Its 24 two-valued states assign equal values to a distinguished pair of distinct rays and hence fail to separate the configuration. The projector cones have 230 facets each. Because the Born table contains genuine $\mathbb{Q}(\sqrt{2})$ entries, the implemented algebraic branch is numerical and returns

$$r_{\Gamma_3} \simeq 0.502197722972, \tag{79}$$

with maximum numerical dual violation $1.6 \times 10^{-10}$. No exact value is claimed for Eq. (79).

### C. Tkadlec's nonunital configuration

Tkadlec's Fig. 2 gives a nonunital Schütte-type configuration in three dimensions [49]. The figure marks 25 rays; the full suborthoposet contains 37 rays in 26 triads. The consolidated audit constructs the full set rather than transcribing it: every orthogonal pair among the 25 marked rays is completed by its unique cross-product ray, and projective duplicates are removed. Independent validation then finds 37 rays, 26 triads, eight two-valued states, and eight rays that are never assigned value one.

The 37-projector cone has 356 facets on each side. The projector LP returns

$$r_{\mathrm{Tkadlec}} \simeq 0.547491649817, \tag{80}$$

with maximum equality residual $2.8 \times 10^{-13}$. No exact primal–dual certificate was recovered, so Eq. (80) is numerical.

### D. Yu–Oh: 13 rays versus 25-ray orthogonal completion

The original Yu–Oh (YO) construction consists of the 13 qutrit rays

$$\begin{aligned}
&(1,0,0),(0,1,0),(0,0,1),\\
&(0,1,1),(0,1,-1),(1,0,1),(1,0,-1),(1,1,0),(1,-1,0),\\
&(1,1,1),(1,1,-1),(1,-1,1),(-1,1,1).
\end{aligned} \tag{81}$$

The projector fragment has 24 cone facets. The simplex linear program gives

$$r_{\mathrm{YO13}} = \frac{3}{8}. \tag{82}$$

In the facet ordering returned by the exact audit program, an optimal certificate can be chosen diagonal, with 18 nonzero diagonal entries and exact residual zero.

The 25-ray version used here is the orthogonal completion of the 13 rays: for every orthogonal pair among the original 13 rays, the cross-product ray completing the triad is added, and duplicates are removed projectively. The completion contains 25 projectors, its cone has 96 facets, and the linear program returns

$$r_{\mathrm{YO25}} = \frac{21}{44} \simeq 0.47727273. \tag{83}$$

The larger value is not surprising: the enlarged projector set generates a different operational cone and supplies more Born constraints to be simulated. The rational audit verifies exact primal and exact dual certificates at $r = 21/44$, so this value is an exact optimum for the restricted projector-cone fragment.

The vector-generated closure audit in Table II makes the cost of the restriction explicit. For the 13-ray set, closing the effects while keeping the original eigenpreparations plus the maximally mixed state gives a numerical optimum near $9/20$. Closing the preparation side as well gives a numerical optimum near $21/44$, equal within the reported numerical precision to the certified value for the 25-projector completion. Equality of these scalar thresholds is reported as a numerical observation, not as a proof that the underlying cones are identical.

### E. Cabello 18–9 and the Peres–Mermin 24–24 completion

The Cabello–Estebaranz–García-Alcaine 18–9 set is a four-dimensional KS configuration with 18 rays in 9 tetrads [13]. The projector-only fragment is represented in the 10-dimensional real symmetric trace basis. The cone generated by the 18 projectors has 146 facets. The numerical optimum rationalizes to $1/3$, and the exact rational audit verifies exact primal and exact dual certificates,

$$r_{18/9} = \frac{1}{3}. \tag{84}$$

so the value is an exact optimum for the restricted projector-cone fragment.

The Peres–Mermin square, analyzed through matrix pencils, yields the 24-24 completion. The 24 vectors include the Cabello 18-ray set as a subset; the remaining rays supply the orthogonal completion associated with the full 24–24 Peres configuration [26]. In the projector-only simplex test, the 24 projectors generate a cone with 120 facets. Writing PM24 for this Peres–Mermin 24-ray fragment, the numerical optimum rationalizes to $4/9$, and the exact rational audit verifies exact primal and exact dual certificates,

$$r_{\mathrm{PM24}} = \frac{4}{9}. \tag{85}$$

The larger value is consistent with adding projectors and hence strengthening the restricted fragment. This row is also an exact optimum for the restricted projector-cone fragment.

TABLE I. Projector-cone depolarizing thresholds for the specified labelled prepare-and-measure fragments. The listed rays are used both as rank-one preparations and rank-one effects, with state-side depolarizing noise $\rho \mapsto \mathrm{tr}(\rho)\mathbb{1}_d/d$. "Exact optimum" means exact primal and dual certificates. "Numerical" means floating-point evidence only. These values are properties of the stated projector-cone fragments and noise map; they are neither invariants of the underlying hypergraphs nor absolute contextuality strengths.

| Configuration | $d$ | projectors | accessible dim. | facets | $r$ | certificate status |
|---|---|---|---|---|---|---|
| Firefly logic $L_{12}$ | 3 | 5 | 4 | 5 | 0 | exact optimum |
| Completed pentagon $C_5$ | 3 | 10 | 6 | 12 | $\approx 0.119140568134$ | numerical/algebraic branch |
| Specker bug | 3 | 13 | 6 | 26 | 25/79 | exact optimum |
| Kochen–Specker $\Gamma_3$ | 3 | 27 | 6 | 230 | $\approx 0.502197722972$ | numerical/algebraic branch |
| Tkadlec Fig. 2, nonunital | 3 | 37 | 6 | 356 | $\approx 0.547491649817$ | numerical |
| Yu–Oh 13-ray projector fragment | 3 | 13 | 6 | 24 | 3/8 | exact optimum |
| Yu–Oh 25-ray orthogonal completion | 3 | 25 | 6 | 96 | 21/44 | exact optimum |
| Cabello–Estebaranz–García-Alcaine 18–9 | 4 | 18 | 10 | 146 | 1/3 | exact optimum |
| Peres–Mermin 24–24 completion | 4 | 24 | 10 | 120 | 4/9 | exact optimum |

### F. Vector-generated operational closures

Table II reports the second audit, in which the same vector sets are closed under the operational data generated by their orthogonality relations. The *closed-effects* rows use the original rank-one eigenpreparations together with the maximally mixed state, and close the effect side under the unit, residual sharp-measurement outcomes, complements, coarse grainings, and equality of operators. The *closed effects+preparations* rows also add normalized versions of the closed effects as preparations. The table does not include the algebraic pentagon, whose exact closure would require an algebraic-number polyhedral backend.

The closed-effects-plus-preparations variant is an additional modelling choice, not a consequence of the original ray hypergraph: it assumes that every normalized closed measurement event is also available as a preparation. It therefore defines a stronger operational fragment than the closed-effects variant.

The closure rows sharpen the comparison with the projector-cone proxy. Firefly remains simplex-embeddable after closure for this specified vector-generated fragment. For Yu–Oh 13 the numerical optimum rationalizes from the exact-certified projector value 3/8 to 9/20 when the effect closure is added, and to 21/44 when the preparation side is closed as well. The 25-ray Yu–Oh completion is unchanged to the reported numerical precision under this closure. Both Specker-bug closures return approximately 25/79, equal within numerical precision to the exact projector-cone optimum. Both $\Gamma_3$ closures return approximately 0.502197722951; their cutting-dual maximum violations are below $7 \times 10^{-15}$. For the Tkadlec nonunital configuration, both closure variants return approximately 0.547491649818, equal within numerical precision to the projector-cone value; the corresponding cutting-dual maximum violations are $7.1 \times 10^{-11}$ and $5.9 \times 10^{-12}$, respectively. Cabello 18–9 has exact-certified projector-cone value 1/3; after effect closure the numerical optimum rationalizes to 4/9, numerically consistent with the Peres–Mermin 24-ray completion. Peres–Mermin/Peres 24 is unchanged to the reported numerical precision at 4/9 under the reported closures.

## XIV. INTERPRETATION OF THE TABLES

Table I answers one narrow question: how much state-side depolarizing noise is needed before the Born bilinear form on the cones generated by the listed projectors admits a simplex embedding? It does not give chromatic numbers, counts of two-valued states, or full operational robustnesses for complete contextuality scenarios. Table II answers a second, more operational question for the source-derived vector rows: what changes when the unit, residual sharp-measurement outcomes, complements, coarse grainings, and simple preparation closures generated by the same vectors are added?

Rows marked "exact optimum" are certified by rational primal and dual witnesses. Rows marked "numerical" or "numerical/algebraic branch" remain floating-point evidence rather than independent exact certificates.

The comparisons should therefore be read as diagnostics rather than rankings. $L_{12}$ is a classically representable partition logic and has $r = 0$ even after closure. The completed pentagon gives a small positive numerical value, as expected for the first cyclic probability-level example. The Specker bug, $\Gamma_3$, and the Tkadlec nonunital configuration exhibit nonfull, nonseparating, and nonunital valuation defects, respectively. The Yu–Oh and Cabello/Peres–Mermin pairs show that orthogonal or operational completion can change the cone and can raise the threshold. The useful content is the uniform computation together with the certificate status of each row.

TABLE II. Vector-generated operational-closure thresholds. Here $(|\Omega|, |E|)$ gives the number of preparation and effect generators after operator equality has been quotiented, and $(f_\Omega, f_E)$ gives the numbers of state- and effect-cone facets. The entries are numerical outputs of the closure script; every entry in the $r$-column is numerical, and displayed fractions are rationalizations of floating-point optima rather than exact values. No row in this table is claimed as an exact-certified optimum.

| Configuration | closure variant | $d$ | $(\|\Omega\|, \|E\|)$ | $(f_\Omega, f_E)$ | solver | $r_{\text{num}}$ | status |
|---|---|---|---|---|---|---|---|
| Firefly logic $L_{12}$ | closed effects | 3 | $(6, 11)$ | $(5, 5)$ | primal | $\approx 0$ | numerical |
| Firefly logic $L_{12}$ | closed effects+preparations | 3 | $(11, 11)$ | $(5, 5)$ | primal | $\approx 0$ | numerical |
| Yu–Oh 13 | closed effects | 3 | $(14, 51)$ | $(24, 96)$ | primal | $\approx 9/20$ | rationalized |
| Yu–Oh 13 | closed effects+preparations | 3 | $(51, 51)$ | $(96, 96)$ | primal | $\approx 21/44$ | rationalized |
| Yu–Oh 25 completion | closed effects | 3 | $(26, 51)$ | $(96, 96)$ | primal | $\approx 21/44$ | rationalized |
| Yu–Oh 25 completion | closed effects+preparations | 3 | $(51, 51)$ | $(96, 96)$ | primal | $\approx 21/44$ | rationalized |
| Specker bug | closed effects | 3 | $(14, 27)$ | $(26, 26)$ | primal | $\approx 25/79$ | rationalized |
| Specker bug | closed effects+preparations | 3 | $(27, 27)$ | $(26, 26)$ | primal | $\approx 25/79$ | rationalized |
| Kochen–Specker $\Gamma_3$ | closed effects | 3 | $(28, 55)$ | $(230, 230)$ | cutting dual | $\approx 0.502197722951$ | numerical |
| Kochen–Specker $\Gamma_3$ | closed effects+preparations | 3 | $(55, 55)$ | $(230, 230)$ | cutting dual | $\approx 0.502197722951$ | numerical |
| Tkadlec Fig. 2, nonunital | closed effects | 3 | $(38, 75)$ | $(356, 356)$ | cutting dual | $\approx 0.547491649818$ | numerical |
| Tkadlec Fig. 2, nonunital | closed effects+preparations | 3 | $(75, 75)$ | $(356, 356)$ | cutting dual | $\approx 0.547491649818$ | numerical |
| Cabello 18–9 | closed effects | 4 | $(19, 121)$ | $(146, 120)$ | primal | $\approx 4/9$ | rationalized |
| Cabello 18–9 | closed effects+preparations | 4 | $(121, 121)$ | $(120, 120)$ | primal | $\approx 4/9$ | rationalized |
| Peres–Mermin/Peres 24 | closed effects | 4 | $(25, 139)$ | $(120, 120)$ | primal | $\approx 4/9$ | rationalized |
| Peres–Mermin/Peres 24 | closed effects+preparations | 4 | $(139, 139)$ | $(120, 120)$ | primal | $\approx 4/9$ | rationalized |

## XV. DISCUSSION

The examples support a modest conclusion. Valuation tests, simplex-embedding tests, and product-rule tests often agree on canonical KS configurations, but they need not expose the same obstruction after the data have been restricted.

This is clearest for operator-valued parity proofs. If an argument is fully resolved into spectral projectors, it may become an ordinary projective quantum-logic problem, as in the Peres–Mermin transcription to the 24-ray completion. If the same argument is kept at the level of coarse-grained commuting observables, the relevant extra assumption may instead be multiplicativity of assigned values. GHZ isolates that point: the collective joint eigensystem is one Boolean block, but the attempted decomposition of its global product observables into context-independent local factors gives the wrong product sign.

The computational tables have the same limited status. Table I compares selected projector cones and reports which values are exactly certified. Table II compares a specified vector-generated operational closure of the source-derived rows. It is not a universal ranking of contextuality. The Specker bug, $\Gamma_3$, and the Tkadlec configuration realize nonfullness, nonseparation, and nonunitality of their two-valued-state spaces; Yu–Oh 13 and Yu–Oh 25, and likewise Cabello 18–9 and Peres–Mermin/Peres 24, generate different cones after orthogonal or operational completion.

Partition logics and chromatic colorability sit naturally on the incidence side of this comparison. Partition logics show that complementarity and non-Boolean pasting do not by themselves force nonclassical probabilities. Strong colorability asks for a global decomposition of the unit assignment into spectral labels, a stricter condition than the mere existence of two-valued states. Both diagnostics are useful, but neither is the same object as a simplex robustness computed from a probability table.

## XVI. CONCLUSION

The useful distinction is therefore not between three new kinds of physics, but between three choices of data: incidence data for valuation, partition-logic, and coloring questions; convex-operational data for simplex-embedding questions; and operator data for functional-composition and product-rule questions.

The resulting distinctions are concrete. A partition-logically representable hypergraph can nevertheless support nonclassical Born probabilities in a quantum realization. Conversely, a closed single Boolean context is simplex-classical. The GHZ contradiction is therefore not located in the collective joint spectral context itself, but in the additional product-preserving identification of its global outcomes with pre-existing local Pauli values. Likewise, simplex nonembeddability, KS uncolorability, and chromatic contextuality diagnose different possible failures of classicality; none is merely a reformulation of the others.

With that restriction in view, the two main contributions are limited but concrete: the GHZ discussion isolates multiplicative product preservation as an additional assumption not contained in the Boolean algebra of the collective joint spectral measurement, and the simplex audits give a uniform set of depolarizing thresholds for both projector-cone fragments and vector-generated closures. The projector-cone rows explicitly marked *exact optimum* in Table I have exact rational primal and dual certificates. The closure rows in Table II show how much the reported thresholds can change once unit effects, residual sharp outcomes, complements, coarse grainings, and normalized event preparations are restored, but they remain numerical unless exact certificates are supplied.

Most importantly, no number in either table is a contextuality strength of a named hypergraph in isolation. Each number belongs to a particular labelled preparation–effect fragment, its specified operational closure (if any), and the chosen state-side depolarizing map. The tables are therefore comparative audits of those fragments, while the incidence, simplex, and product-rule tests remain distinct diagnostics of different retained structures.

## DATA AND CODE AVAILABILITY

The supplementary archive, available at `https://svozil.github.io/publications/2026-simplex-simplex_supplement_bundle.zip`, contains one self-contained program, `simplex_audit_rebuilt.py`, the complete `audit_v3` output directory, the explicit vector-and-context catalogue, the supplemental text, and SHA-256 checksums. The archived program hash is `d2298aeaab2d47a42e94e9077ece208df8042ddf6603d6fd0252d8fc184e1596`; the same hash is recorded by the run manifest. Exact regeneration of rational facets and certificates requires a fraction/GMP-enabled `pycddlib`/cddlib installation. The default command `python simplex_audit_rebuilt.py --outdir audit_v3` validates every source configuration, runs every projector and closure task, and reuses a task only when its input fingerprint matches. Software versions are recorded in the `runtime` object of `audit_v3/audit_report.json`.

## ACKNOWLEDGMENTS

This research was funded in whole or in part by the *Austrian Science Fund (FWF)* [Grant digital object identifier (DOI): 10.55776/PIN5424624]. The author acknowledges TU Wien Bibliothek for financial support through its Open Access Funding Programme.

OpenAI Codex (GPT-5.6) was used to assist with manuscript criticism, literature organization, checking derivations, and editorial revision. The author directed its use, independently verified the mathematical arguments and cited sources, revised the resulting text, and assumes full responsibility for the manuscript.

## Appendix A: Computational audit data

This appendix records the computational status behind Tables I and II. It is included because the distinction between numerical optima, exact primal certificates, and exact dual certificates is essential for interpreting the tables.

The exact audit uses the accessible-fragment form of Eq. (75). Integer or rational rays are converted to raw real-symmetric projector coordinates. Effects are stored in dual coordinates, $\widetilde{I}_E = GI_E$, where $G = \mathrm{diag}(\underbrace{1,\dots,1}_{d},\underbrace{2,\dots,2}_{d(d-1)/2})$, and cddlib is used over $\mathbb{Q}$ to obtain rational facet matrices whenever possible. A numerical linear program first proposes $r$ and $\sigma$. The proposed rational values are then checked exactly by verifying

$$H_E^T \sigma H_\Omega = \widetilde{I}_E^T \big[(1-r)\,\mathrm{Id}_{\mathcal{A}} + rD\big] I_\Omega, \qquad \sigma \geq_{\mathrm{e}} 0. \quad \text{(A1)}$$

This is a primal feasibility certificate. Exact optimality additionally requires a rational dual witness. The audit verifies such dual witnesses for all rational rows reported as exact optima in Table I.

For the Specker bug, the ray representatives lie in $\mathbb{Q}(\sqrt{2})$ but all projector overlaps are rational. The program rank-factorizes that Born table over $\mathbb{Q}$ and applies the same cone-facet and primal–dual checks in the resulting six-dimensional operational coordinates. For $\Gamma_3$, some Born overlaps remain in $\mathbb{Q}(\sqrt{2}) \setminus \mathbb{Q}$; its operational rank factorization, hull facets, and LP are therefore numerical.

The dual certificate is as follows. Put

$$M_0 = \widetilde{I}_E^T I_\Omega, \qquad M_1 = \widetilde{I}_E^T (D - \mathrm{Id}_{\mathcal{A}}) I_\Omega, \quad \text{(A2)}$$

and let $C_{\ell k} = (h_\ell^E)^T h_k^\Omega$ denote the rank-one coefficient matrix multiplying $\sigma_{\ell k}$. Omitting the inactive upper bound $r \leq 1$ gives the lower-bounded primal LP

$$\begin{aligned} &\min\ r \\ &\text{s.t.}\quad \sum_{\ell,k} \sigma_{\ell k} C_{\ell k} - rM_1 = M_0, \\ &\qquad\quad \sigma_{\ell k} \geq 0, \qquad r \geq 0. \end{aligned} \quad \text{(A3)}$$

All reported certified optima satisfy $0 \leq r \leq 1$, so a certificate for Eq. (A3), together with primal feasibility for the bounded problem, certifies the implemented LP. With the Frobenius inner product $\langle A, B\rangle = \mathrm{tr}(A^T B)$ on the accessible coefficient matrices, the dual is

$$\begin{aligned} &\max\ \langle Y, M_0\rangle \\ &\text{s.t.}\quad \langle Y, C_{\ell k}\rangle \leq 0 \quad \text{for all } \ell, k, \\ &\qquad\quad -\langle Y, M_1\rangle \leq 1. \end{aligned} \quad \text{(A4)}$$

A rational matrix $Y$ satisfying these inequalities and $\langle Y, M_0\rangle = r_*$ proves that no feasible point has $r < r_*$. Together with a rational nonnegative $\sigma$ satisfying Eq. (A1) at $r_*$, this is the primal–dual certificate used for the "exact optimum" rows.

The projector-cone Python audit returned the status summarized in Table III. The table is intentionally diagnostic rather than a second robustness table: its purpose is to show which entries are exact optima and which remain numerical. For each rational row marked "exact optimum," exact primal feasibility is checked by Eq. (A1), and exact optimality is certified by a rational dual witness recovered from the numerical dual active set when direct rationalization is insufficient.

The completed pentagon is not exact-certified by the rational backend because the coordinates in Eq. (51) are algebraic. The numerical branch found 10 projectors, full accessible dimension six, 12 facets, numerical robustness $r = 0.119140568134$, maximum residual approximately $1.8 \times 10^{-13}$, and 16 active $\sigma$ entries. The $\Gamma_3$ branch found 27 projectors, 230 facets per cone, and $r = 0.502197722972$, with maximum dual violation approximately $1.6 \times 10^{-10}$. The rational Tkadlec nonunital input has exact facets, but its optimum was not recovered as an exact primal–dual pair; its displayed value is therefore numerical. Exact certification of the pentagon and $\Gamma_3$ would require an algebraic-number polyhedral backend or a rationally equivalent reformulation.

The same program performed the vector-closure audit. For each source-derived vector set it generated residual sharp-measurement effects, complements, coarse grainings, and duplicate-operator quotients from the vector data itself. The run reported in Table II used exact cddlib facets throughout the rational closure rows. The large $\Gamma_3$ and Tkadlec rows were solved by a cutting-dual linear program; this is numerically much smaller than the primal $\sigma$-program but does not return an exact primal certificate. Consequently every value in Table II is reported as numerical. The complete command `python simplex_audit_rebuilt.py --outdir audit_v3` finished without error and wrote the archived manifests.


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TABLE III. Audit status of the simplex computations. Here $k_\Omega$ and $k_E$ are the accessible state and effect dimensions after Stage-1 reduction. The rational rows marked “exact optimum” have both exact primal and exact dual certificates.

| Configuration | projectors | $k_\Omega$ | $k_E$ | facets | $r$ | exact primal | exact dual | note |
|---|---|---|---|---|---|---|---|---|
| Firefly logic $L_{12}$ | 5 | 4 | 4 | 5 | 0 | yes | yes | exact optimum |
| Yu–Oh 13 | 13 | 6 | 6 | 24 | 3/8 | yes | yes | exact optimum |
| Yu–Oh 25 completion | 25 | 6 | 6 | 96 | 21/44 | yes | yes | exact optimum |
| Specker bug | 13 | 6 | 6 | 26 | 25/79 | yes | yes | exact optimum |
| Kochen–Specker $\Gamma_3$ | 27 | 6 | 6 | 230 | $\approx 0.502197722972$ | no | no | algebraic/numerical |
| Tkadlec Fig. 2, nonunital | 37 | 6 | 6 | 356 | $\approx 0.547491649817$ | no | no | numerical |
| Cabello 18–9 | 18 | 10 | 10 | 146 | 1/3 | yes | yes | exact optimum |
| Peres–Mermin/Peres 24 | 24 | 10 | 10 | 120 | 4/9 | yes | yes | exact optimum |
| Completed pentagon $C_5$ | 10 | 6 | 6 | 12 | $\approx 0.119140568134$ | no | no | algebraic/numerical branch |


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